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SESSION TOPIC: CAPITAL ASSET PRICING THEORY

SESSION CHAIRPERSON: MARTIN J. GRUBER

THE CURRENT STATUS OF THE CAPITAL ASSET PRICING MODEL (CAPM)

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INTRODUCTION

THIS IS A BIASED REPORT on the status of a paradigm, the mean-variance Capital Asset Pricing Model, *the* CAPM. The report is not exhaustive—as a survey would be—and highlights areas of development which seem to me to be of most importance. It tries, however, to stick reasonably closely to its subject matter, (even at the expense of topics which are of greater interest to me), and I apologize at the outset to anyone who feels slighted if their taste in these matters is not coincident with my own.

The report will also have a distinctly positive focus; the attractiveness of the CAPM is due to its potential testability. It is a paradigm, precisely because it is cast in terms of variables which are, at least in principle and with the usual exception of the ex ante-ex post distinction, empirically observable and statistically testable. Its positive orientation and apparently simple intuition have made it the central equilibrium model of financial economics, and it is definitely not, as some have suggested, merely a particular example or parametrization of a rather simple general equilibrium model. On the other hand, it is also not, as some enthusiasts believe, the only or merely the 'best' possible model. There is, in fact, an alternative model, the arbitrage pricing theory (APT for short) that appears to offer the hope of retaining the simplicity of the CAPM and its positive empirical orientation while avoiding many of the theoretical difficulties and empirical problems that we will describe in this report.

The following procession outlines the basic structure of the CAPM:

First, for any of several reasons individual investors choose portfolios that they perceive to be mean variance efficient.

Second, all investors have the same ex ante beliefs. In particular, they share a common view of the expected return, covariance pair, (E, V).

Third, the market portfolio, m, is a convex combination of the individual portfolios. Since the set of efficient portfolios for a given (E, V) pair is convex, it follows that the market portfolio is mean variance efficient.

Fourth, an equivalent expression of the mean variance efficiency of the market

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portfolio is that the risk premium for any asset i is linearly related to its covariance with the market portfolio,

$$\beta_i \equiv \frac{\sigma_{im}^2}{\sigma_m^2}$$

and

$$E_i - E_0 = \beta_i (E_m - E_0), \tag{1}$$

where E denotes expected return and E_0 is the return on a portfolio uncorrelated with the market portfolio, the riskless asset if one exists.

Fifth, testing the theory relies on the assumption that the ex post distribution from which returns are drawn is the ex ante one perceived by investors. It follows from multivariate normality that (1) directly satisfies the Gauss-Markov regression assumptions where

$$\tilde{X}_i = E_0 + \beta_i (\tilde{X}_m - E_0) + \tilde{\epsilon}_i \,, \tag{2}$$

and

$$\text{cov}\,(\tilde{\epsilon}_i, \tilde{X}_m) = 0 \,.$$

Testing proceeds, using the regression of (2), by making stationarity assumptions on the intertemporal distribution of returns. Although the theory is static, stationarity (or some explicit model of nonstationarity) is required to provide a sufficient replication of returns for statistical verification.

Sixth, attempts have been made to extend the model to richer environments of choice and returns, such as intertemporal frameworks, and to analyses which take account of institutional characteristics such as taxation.

The CAPM theory outlined above not only "explains" asset prices, but it does so by providing an analytic basis for a brilliant, if not entirely reliable, intuition. Asset risk premia depend not on the total risk of the asset, but rather on the relationship of the asset to the overall market. Since in aggregate all risk is borne by the market portfolio, only the relationship between the asset and the market portfolio, its beta, can determine the premium for an individual asset. The organization of this modest report follows the above outline of development of the model, but since each wrinkle at any step has implications for the whole model, we will explore points beyond the outline.

1. Assumptions on Choice

The basic foundations of the theory were laid down in the microeconomic studies of mean variance choice by Markowitz [1959] and Tobin [1958]. The critical extension to an equilibrium in the capital market, and the development of the CAPM, was accomplished by Sharpe [1964] and Lintner [1965]. The CAPM substantiated the idea that, in competitive equilibrium, assets earn premia over the riskless rate that increase with their risk, by showing that the determining influence on risk premia is the covariance between the asset and the market portfolio, rather

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than the own or intrinsic risk of the asset. From an economic perspective risk is priced through its relationship or substitutability with other assets and not on its own. The second and somewhat surprising conclusion that emerged from the analysis was the separation theorem. In a mean variance world, each individual agent finds it optimal to hold a portfolio that is a linear combination of a riskless asset and a single risky portfolio. The composition of this portfolio is independent of risk preferences, or those parameters of preference which are permitted to vary; such choice variations enter in only to determine the allocation of wealth between the riskless asset and the risky portfolio. Black [1972] extended this analysis to a world without a riskless asset and showed that two funds of risky assets could be identified which span the efficient portfolio set. (See Long [1972] for an examination of the properties of Black's zero beta portfolio.)

These two results, the risk premia analysis on the one hand and the separation argument on the other, engendered a bit of confusion, and the search for conditions on either choice or technology which would guarantee the first, became more a search for conditions which led to separation. The two arguments are related but their relationship hinges on which of two routes is taken in developing the analysis.

To make empirically meaningful statements about the general equilibrium pattern of competitive returns requires restrictions on either technology, or choice, or on the pair. Restrictions on preferences which lead to separation results were exhaustively studied by Cass and Stiglitz [1970], building on earlier work by Hakansson [1969], Leland [1971] and Mossin [1968]. Cass and Stiglitz found necessary and sufficient conditions on investor utility functions for investors to have separating optimal portfolio decisions. Utility functions in the Cass-Stiglitz class have the property that the associated optimal portfolios formed from n assets can be grouped into $k < n$ funds. The allocation of wealth to the k individual funds may change with the investor's wealth, but the composition of the funds will not. This work generalized the earlier observation that constant relative risk aversion utility functions were necessary and sufficient for the investor to have one fund separation, i.e., a portfolio whose composition is wealth independent; and it also generalized the Markowitz and Tobin analyses which showed that a quadratic utility function would imply that wealth would be divided between a single fund of risky assets and a riskless asset.

Finding utility or preference based separation, though, is not the same thing as finding conditions which lead to the two-parameter capital asset pricing model. If, for example, all investors choose the same single portfolio of risky assets, then in equilibrium that portfolio must be the market portfolio, but there is no need for it to be on the mean-variance efficient frontier and, as a consequence, the CAPM as we usually think of it need not be valid. Of course, the market portfolio will be efficient with regard to the utility class that led to separation and that is potentially testable, but it is a different theory from the CAPM and not terribly much has been done in this direction.[1]

1. Building on Hakansson's work, Rubinstein [1974], however, has examined an equilibrium model for logarithmic utility, and has further studied financial theory from this perspective. Roll [1973] has

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I believe there is a sound reason why explorations of utility-based CAPM theories have not attracted more effort. In any such theory, when we aggregate to the market level we must impose severe homogeneity requirements on utility functions to get results that limit data. In addition, we are also forced to impose strong homogeneity restrictions on perceived distributions; first, to keep the results tractible and, second, and most importantly, to relate the theory to observables. If individuals perceive nontrivially distinct ex ante patterns of returns, then what are we to make of the relationship between the ex post returns and the ex ante beliefs? Any theory that purports to be testable must establish such a relationship and it is most straightforward to require strong homogeneity of individual beliefs. Given that such homogeneity is going to be imposed eventually, it would seem natural to begin the CAPM story with restrictions on distributions, rather than preferences. A theory that obtains strong implications for equilibrium asset prices from restrictions on perceived distributions and permits heterogeneity in preferences is surely to be preferrred to one which obtains similar market implications, but imposes restrictions on preferences along with strong similarity of beliefs.

Markowitz and Tobin noted that the multivariate normal distribution was a sufficient condition for asset choice to satisfy two fund separation. Since it is also a sufficient condition for all risk averse investors to choose mean-variance efficient portfolios, as Sharpe and Lintner observed, it could be the basis for the CAPM. Tobin also remarked that any such "two parameter" distribution would be sufficient; one parameter for expected return and another, variance, as a measure of risk. As a number of subsequent writers observed, this was not quite the case, but some, in particular Fama [1968], did extend the conditions to versions of the multivariate stable class.[2]

The full set of necessary and sufficient conditions for distributional separation, i.e., the existence of $k < n$ funds of assets such that *any* risk averse investor would choose a portfolio that is simply a combination of the k funds, were found in Ross [1978*a*]. The separating distributions are really types of linear factor models and they stress that separation depends primarily on the corelations amongst asset returns and not on their particular marginal returns, e.g., normal or stable.[3] Furthermore, when there is $(k=2)$ two fund separation, then even though the distributions might differ significantly from the normal, when variances exist the efficient frontier will *coincide* with the mean variance efficient frontier! In other words, two fund separation based on distributional restrictions is actually sufficient and, in an important sense, necessary for the mean variance CAPM to hold.

Further developments in this area would seem to lie in two directions. First, we do not have any work at all on paired restrictions that lead to a CAPM. We have

also developed such a model and done some empirical work on it. In addition, there is a considerable literature on expanding the utility function beyond its second moment in a Taylor series and examining skew effects (see e.g., Kraus and Litzenberger [1976] or Rubinstein [1973a]). These efforts seem to have diminished in recent years.

2. See Bawa [1978] for a recent extension to the Student class of distributions.

3. Although, when we add in some independence conditions the stables regain their unique necessary role (see Ross [1978b]).

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examined preference restrictions and now we have the distributional restrictions that 'work' for any preferences, but there are *no extensions* of the sort "this distribution class and this (limited) preference class work". Such analysis will be very difficult and I am not optimistic about its possibilities, but success in this direction would greatly enhance our understanding of the structure of portfolio problems.

A second and more promising route lies in the direction of approximate results. In large markets some complications that prevent exact separation may wash out or be diversified away in efficient portfolios. This sort of thinking leads naturally to the arbitrage model of asset pricing, APT (See Ross [1976 a and b]). We will defer looking at this alternative to the CAPM until the section on empirical testing.

2. Homogeneity of Beliefs

The question of what investors do and do not know and the relationship of this information to the CAPM has been one of the most active areas of research in financial economics. A review of the efficient markets literature would be a whole study on its own (see Fama [1970]), and we will concentrate only on some aspects that seem most pertinent for the CAPM.

As we have seen, the CAPM is based on the assumption that investors share a common ex ante distribution that they perceive generates returns. To test the theory, then, it is natural to assume that investors do not err in a systematic fashion in their a priori beliefs. It follows that the ex post distribution from which returns are drawn will be the ex ante one on which investors based their actions. In particular, given any preselected information set, e.g., the past price observations, expectations conditional on this set are assumed to coincide with ex post population means and this assumption is jointly tested with the CAPM. It is this linkage between ex ante beliefs and observables that has forced us into the assumption that investors hold similar beliefs, and not any inability to derive formulas that describe equilibrium returns as some function of the various ex ante views.

Of course, casual observation suggests quite the opposite; it seems clear that investors do not show the homogeneity of beliefs which characterize our theories. To the contrary, it is widely believed that it is differences in beliefs and disparate changes in these beliefs which occasion trade, and there have been a number of efforts directed at modelling such differences and assessing their impact on asset pricing. Epps [1976] and Copeland [1976], for example, have examined models where prices are determined by the interaction of individual excess demand functions in an asset market, and random shocks to these curves generate random movements in equilibrium prices.[4]

Aside from their somewhat ad hoc formulation the common difficulty with these analyses is that they imply a persistent irrationality on the part of investors. In all of these models the ex post distribution of returns differs in a systematic and predictable fashion from the ex ante ones upon which investors based their decisions (in some, the ex ante beliefs are not even specified and only an excess

4. Miller [1977] has argued that, with short sales restrictions, stock prices tend to overvalue true returns since individual stocks will be held by investors who, on average, overestimate them.

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demand function is assumed). It would not be difficult to incorporate some random cross sectional heterogeneity in investor beliefs into the CAPM, but since a single ex post distribution of returns is observed by all, over time one would not expect to observe systematic and persistent differences.[5]

A number of researchers have examined the rationality issue in capital markets paying special attention to this relationship between prices and beliefs. In their models, not only do prices clear markets for any given set of ex ante beliefs on future performance, they also enter individual information sets as datum. This simultaneity can lead to paradoxes.

The simplest sort of paradox arises when there are two types of individuals, informed and uninformed. If the equilibrium price in such a world is sufficient to convey the information possessed by the informed group, then upon observing the price the uninformed group will be informed. With all parties informed, though, a new equilibrium price will be established and this new price may *not* convey the requisite information. The problem with this approach to equilibrium is that the single price (function) must not only clear markets, it must also do so conditional on the information which it conveys (see Green [1973], Kihlstrom and Mirman [1975], Hart [1975], Grossman [1977], Jordan [1976] or Kreps [1977]). Furthermore, if the informed must bear a cost to acquire their original information, while the uninformed can acquire the same information by freely observing the market price, then this inability to appropriate the gains from information acquisition will lead to a collapse of such activities. On the other hand, given that no one is acquiring information, it may pay one party to do so (see Grossman and Stiglitz [1976]). This lack of equilibrium cuts to the heart of our understanding of the role of information in capital asset pricing. Grossman [1976], for example, has displayed a simple mean variance model where current equilibrium prices are a sufficient statistic for the special information possessed by the participants and where, hence, the above paradoxes can arise.[6]

On a somewhat more macroscopic level, some work on information has been done within the framework of the intertemporal rational expectations literature. We will take this up in detail below, but for the moment we should explain that the bulk of this work is carried out under a representative individual assumption, i.e., the "market" possesses the relevant information. Individuals may differ along any of the dimensions discussed in Section 1 that still permit the aggregate CAPM to

5. This view is based on a Bayesian argument that eventually, with diffuse priors, the ex post observational data will swamp the priors and produce similar posteriors. Even with random news received idiosyncratically there would not be persistent differences in any one direction for any one agent. Unfortunately, this intuition is based on the assumption that the economic data follows a stationary process and this need not be the case when intertemporal prices are generated as a sequence of temporary equilibria. Jordan [1976] has produced a counterexample of this sort in which posteriori distributions do not settle down to any nice path. Without stationarity though, or some explicit model of nonstationarity, econometrics itself is in jeopardy and this seems too tragic to take seriously.

6. Grossman [1977] further argues that with enough noise (e.g., in supply) this sufficiency disappears, and concludes that markets which are too informationally perfect must close down leaving only those where there is enough noise to vitiate the sufficiency of equilibrium prices. Over time, the nonstationarity of Jordan's example might prove necessary to prevent this sort of breakdown when markets are too informationally perfect.

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hold, but they do not have nontrivially distinct information sets.[7] If they did differ sufficiently, leaving aside questions of whether prices transmit information, arbitrage problems can threaten the existence of equilibrium (see Green [1974]).

The work on information is highly provocative, but definitive positive conclusions for asset pricing have yet to emerge. In particular, we are still groping for an adequate understanding of exactly how competitive prices are determined in such markets with asymmetric information and at present there are several competing conceptions of what constitutes an equilibrium. A resolution of these issues is going to turn on a closer examination of the microtheory of price formation in these markets, and that is a task that is just beginning.

3. Aggregation to the Market Level

There is not really very much that has to be said about this step. There are some issues of existence of equilibrium that linger since, without short sales restrictions, the budget sets of individuals do not possess the compactness property used in general equilibrium existence proofs, but these were largely resolved by Hart [1974].

The key property that permits aggregation is the convexity of the efficient set of portfolios. Since the market portfolio is a convex combination of individual portfolios, if they are individually efficient, then the market must also be efficient. How general is this property? To put the matter a bit differently, will the efficient set of portfolios associated with any distribution be convex?

Unfortunately, outside of complete markets, the answer is in the negative. Dybvig and Ross [1977] have constructed counterexamples where the efficient frontier is not convex, and whether the class of returns for which the frontier is convex can be meaningfully extended beyond the Arrow-Debreu case (and more than two assets) is also not known.

4. and 5. Testing the CAPM

Undoubtably the most exciting work on the CAPM in recent years has occurred on the empirical front. In a major study Roll [1977] has called into question the vast mountain of empirical tests of this theory and, at the least, has forced a reevaluation and reinterpretation of such tests. There are two central elements to Roll's critique.

First, the CAPM as we have developed it above, implies that the market portfolio, m, is mean variance efficient. Furthermore, other implications of the CAPM, notably the linearity of the risk premia expression (1) in beta, are actually equivalent to the efficiency of the market portfolio. In other words, expected returns are linear in beta if and only if the market portfolio is mean variance

7. Another set of issues related to the informational ones concerns the workings of incomplete markets. The questions associated with the Pareto efficiency of the CAPM when it is used as an investment criteria, the unanimity literature (see, e.g., Stiglitz [1972], Jensen and Long [1972], Merton and Subrahmanyam [1974], Leland [1974], Radner [1974] or Ekern and Wilson [1974]), or the results of Hart [1975] and Satterthwaite [1976] are all ignored in this paper in deference to our narrow positivist focus. In addition, as is now apparent from the separation results, the unanimity objections when applied to the CAPM are really assertions of non price taking behavior and, as an empirical matter, it is difficult to think that they have much force.

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efficient (see Ross [1977]). As a consequence the CAPM can only be refuted by examining implications of the statement that the market portfolio is mean-variance efficient. Unfortunately, though, this equivalence has played little if any role in previous tests of the CAPM.

The bulk of the tests (Jensen [1972], Black, Jensen and Scholes [1972], Blume and Friend [1973], or Fama and MacBeth [1973] to mention a few) rely on explicit proxies for the market portfolio. The commonest such proxies are the value weighted, and the equal weighted portfolio, about which, unfortunately, the CAPM has absolutely nothing to say. The equal weighted portfolio may or may not be efficient, but even if it is, such a finding would have no implications whatsoever for acceptance or rejection of the CAPM theory (see Roll [1977] or Ross [1976a]). This is an unfortunate state of affairs; the market portfolio itself is much harder to observe in practice than in principle. Certainly, restricting the portfolio to the issues listed on the NYSE, and even using a value weighted portfolio, restricts it to a small portion of the totality of traded securities, but it seems that for empirical work we are going to have to settle for looking at subsets of the whole set of assets. Fama, in particular, seems to be aware of this difficulty and summarizes the current state of empirical work by the observation that "In truth, all we can really say at this time is that the literature has not yet produced a meaningful test of the Sharpe-Lintner hypothesis."[8] Suppose, though, that we ignore all of these issues and agree that the proposition we wish to test is the (ex ante) efficiency of a proxy portfolio, p.[9]

Roll's second point is somewhat subtler and equally damaging. Suppose that the market portfolio or whatever proxy, p, we are using is *not*, in fact, ex ante efficient. Letting β denote the vector of regression coefficients with this portfolio, we have

$$E_i - E_0 = (E_p - E_0)\,\beta_i + \alpha_i, \tag{3}$$

where the equation simply defines α_i. If for all i, $\alpha_i = 0$, then p is efficient, but suppose that it is not. Since the proxy will have positive amounts in all assets we know that a weighted average of α_i, with the weights using the proportions in p, averages the α_i deviations to zero. Now, a cross sectional regression over assets will appear significant if α_i and β_i are unrelated. In general, in a finite sample, the regression deviations can be attributed either to ex ante inefficiency or to sampling variation. Furthermore, grouping procedures will tend to increase the significance of the regression. In other words, there is a whole family of inefficient portfolios for which the usual statistical tests of the linear relation (3) will falsely accept the efficiency of the proxy (since the alternative hypothesis of inefficiency is dense in the null under these tests, they have very low power). Of course, with an arbitrarily large time series sample and assuming that E and V, the covariance matrix, are constant over time, asymptotically we will be able to test (3) exactly, but this best of all worlds is of little relevance for actual testing. Notice, too, that ex post, i.e.,

8. Fama [1976, p. 370]. Somewhat schizophrenically, though, the sentence before this one asserts that "The overall view of market equilibrium is that of the Sharpe-Lintner model."

9. As Roll [1977] points out, tests which rely on the supposed closeness of m and p can be statistically misleading since the efficiency characteristics of close portfolios can be quite different.

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given the sample, there exists a whole efficient set of proxies, p, which satisfy (3) with $\alpha_i = 0$; as a consequence, much hinges on our faith in the ex ante choice of p.

The response to Roll's critique has been largely a rejection of what is misinterpreted as a nihilistic message. "If we take the Roll objections to heart, then no theory is testable."[10] This pessimism is not wholly warranted. Roll's analysis does *not* say that we must observe m to test the theory. For example, we can all agree that m has only positive components, and this means that the CAPM could be refuted by refuting the existence of a positive (ex ante) efficient portfolio. This is a testable proposition, but as Roll's second point shows, the construction of suitably powerful tests will be very difficult. More generally, even though m is unobservable, a proxy p can be used for testing if we know something about $m - p$. By bounding the difference between m and p—for instance, by bounding the total wealth not included in our sample—it might be possible to construct tests of the efficiency of m by using p. This will not be easy, but it should be possible.

Furthermore, it is indefensible to argue that if m is unobservable, "then no theory is testable." The arbitrage theory of capital asset pricing (APT), while conceptually different, retains the desirable simple features of the original theory (see Ross [1976a and b]). The APT begins with a generalization of the separating distributions, which are themselves generalizations of Arrow-Debreu tableaus of less than full rank, and from that framework develops a consistent and testable theory of capital asset pricing. For illustration, suppose that returns are generated by a two factor model, say

$$\tilde{R}_i = E_i + \beta_{i1}\,\tilde{R}_{p1} + \beta_{i2}\,\tilde{R}_{p2} + \tilde{\epsilon}_i, \tag{4}$$

where $\tilde{\epsilon}_i$ is a noise term independent of the returns of the two factors, $\tilde{R}_{p1}$ and $\tilde{R}_{p2}$, and of $\tilde{\epsilon}_j$. The identify of $\tilde{R}_{p1}$ and $\tilde{R}_{p2}$ is of great importance for economics, but not for the argument which follows. With a large number of assets all well diversified portfolios will diversify away the noise terms, $\tilde{\epsilon}$, by the law of large numbers, and only the systematic risks of holding $\tilde{R}_{p1}$ and $\tilde{R}_{p2}$ will remain. In other words, the returns on such portfolios, a, will be given by

$$\tilde{R}_a \approx E_a + \beta_{a1}\,\tilde{R}_{p1} + \beta_{a2}\,\tilde{R}_{p2}. \tag{5}$$

To prevent arbitrage possibilities in such a world it must be the case that E_a and β_{a1} and β_{a2} are linearly related for all a, and if E_0 is the return on an asset with no systematic risk, e.g., a riskless asset, then

$$E_a - E_0 \cong (E_{p1} - E_0)\,\beta_{a1} + (E_{p2} - E_0)\,\beta_{a2}. \tag{6}$$

This relationship accords no special role to the market portfolio, makes no assumptions on the normality of returns, only on their corelations as in (4), and is testable in subsets of the data over which (4) is presumed to hold. Unlike the CAPM, APT also generalizes to intertemporal models. In addition, it is consistent with, for example, the modern theory of option pricing. The arguments which lead to a hedge portfolio (see Black and Scholes [1973] or Cox and Ross [1976]) are

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precisely the ones we have employed when $\tilde{\epsilon}_i = 0$ to permit exact hedging and when there is a single factor, the returns on the underlying stock.

Testing the CAPM or any other theory, for that matter, cannot be done in isolation; there must be a viable alternative to the theory under discussion. If there is a single strong intuition that underlies modern finance it is the belief that there are many close substitutes in financial markets and that such substitutes will be priced together. This is the intuition of the APT and it is what makes it testable on subsets. Ideally, we would like to test the CAPM directly against the APT, but unfortunately the empirical problems of the CAPM make it quite difficult to specify such tests. On the other hand, if the problems raised by Roll prove insurmountable, we may never be able to construct a robust test to reject the CAPM. If the APT theory is not rejected in testing, then it will be the natural successor to the CAPM.[11]

6. Extensions of the CAPM

A. *Inflation, International Finance, Human Capital, Taxation, etc.*

There have been numerous extensions of the basic theory to cover a variety of situations. In nearly all of these the CAPM ceases to be valid, i.e., it is no longer true that the market portfolio is mean variance efficient. It is not possible to treat these exhaustively, however we can stress those efforts which have required modifications of the original model rather than simple applications.

First, a body of literature has developed which extends the model to the international economy and, closely related efforts have considered the impact of inflation on returns. At a fundamental level this work is really a modification of the original choice set and a discussion of it rightfully belongs to Section 1. Solnik [1974] was the first to develop an international capital asset pricing model, and Roll and Solnik [1975], Grauer, Litzenberger and Stohle [1976], and others have extended this work and done empirical testing.

The basic argument is that returns are measured in a nominal accounting unit, say dollars, while preference is over real goods. This requires deflating returns by a price index to translate them into real returns. The CAPM given above now holds identically as before, with the understanding that all returns must be evaluated in real terms. Once again the CAPM is equivalent to the real efficiency of the market portfolio. (Similarly the entire analysis can be carried out in a straightforward fashion within the context of the APT.) Of course, this abstracts from all of the usual problems with the construction of price indices, and with many real goods we must make homotheticity assumptions on preferences to insure the stability of the index over any range of returns. If the data is expressed in nominal terms, then we can rewrite the real CAPM market line equation in nominal terms by the addition of a term to represent covariances between securities and the price index—this term will influence the nominal risk premium.

The problem of international finance entails trade among groups of individuals with different national currencies. What is relevant for equilibrium pricing, though,

11. It can also be argued that much of the spurious past successes in testing the CAPM actually were tests of the APT theory.

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is that individuals possess different price indices and there is every reason to believe that individuals can be as heterogeneous in their tastes within a country as across countries. Leaving these issues aside, if we identify indices with currencies, then we have a particular form of heterogeneity of preferences that modifies the previous analysis; now there is no single identifiable index to use to measure 'real' returns. What is clear is that in terms of any particular index, the aggregate portfolio including both domestic and foreign holdings by individuals with that index will be real efficient. By expressing all returns in terms of any single numeraire currency, an aggregate expression can be derived for risk premia in terms of covariance with the world portfolio and covariance with a bond portfolio that hedges against exchange risk. (See Ross and Walsh [1977] for a development of this approach.)

In one form or another, this mode of analysis is common to all of the studies in this area and it is fair to say that they have increased our understanding of pricing in a world of uncertain inflation and where capital moves freely across international boundaries. (See Black [1974] for an examination of the effect of tariff barriers on these analyses.) But, the major work remains. How are these analyses to be integrated into the framework of international trade? What is the relationship between real capital flows and the financial equilibrium relationships and what, if any, are the repercussions for international economics?[12] Can risk serve to insulate a country's interest rates to a sufficient extent so as to permit them to pursue independent monetary policies, or will such attempts always show up in divergent exchange rate expectations or in changes in foreign reserves? These issues are untouched and they are the critical questions that must soon be addressed by those who wish to apply the CAPM methodology and intuition to international finance.

The CAPM has also been used to examine the effects of non-marketed assets, such as human capital, on pricing. Mayers [1972, 1973] who first looked at this extension, finds that with an exogenous supply of human capital, asset prices and risk premia will be, in part, determined by covariances of returns with the returns on human capital. The analytics of this argument are identical to those of the inflation argument. This can be seen easily by noting that (at least in a naive diffusion model) real returns are nominal returns minus the deflation and, similarly, with non traded human capital, returns are augmented by the returns on such nonmarketed assets. Just as, ceteris paribus, assets are used to hedge against inflation, portfolios are also chosen to hedge against adverse movements in the returns on human capital. But, the central problems within this field remain. The peculiarity of human capital lies in both its nonmarketability and its intertemporal properties of irreversibility. To date the extremely difficult questions posed by the interplay of both of these features remain unresolved. In addition, the need to integrate this work into labor market theory and, in particular, into the controversy between the training and marginal productivity theories of wage determination on the one hand, and the signalling literature on the other, awaits a deft hand.

As for the empirical testing of these extensions suffice it to say that at present it is plagued by all of the problems inherent in testing the original model and some

12. The first part of this question arises in the closed model as well.

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special ones of its own. In testing human capital formulations of the CAPM, for example, the measurement of the return to human capital is so difficult, I suspect that rather than 'testing' the CAPM we should just use the APT version to develop data on such returns as a residual (which requires us to separate out such fundamentally indeterminate quantities as monopoly rents).[13]

A second area of development deals with the impact of taxation on capital asset pricing. The study of corporate taxation and its relationship to financial equilibrium is a rapidly developing area of research, certainly in large measure spurred by Merton Miller's [1977] provocative AFA presidential address in which he argued that in a general equilibrium with a graduated personal tax structure, aggregate corporate debt would be determined, but individual corporations would derive no net tax benefits from debt financing. A similar type of negative result was exploited by Black and Scholes [1974] in their analysis of the tax effects of dividend yield. Brennan [1970] and Long [1977] have developed the analysis of the impact of differential taxation of capital gains and dividends on the efficient set mean variance analysis.

These papers, though, are more by way of exceptions than the rule and there has not been much work done on the precise impact of taxation on pricing in the CAPM framework. Of course, all of the current work on taxation influences the CAPM, but in line with our narrow charge it lies outside the scope of the present paper.

B. *Intertemporal models*

The CAPM is a static one period model and a considerable effort has been directed at intertemporal extensions. The most successful of these have been those based on Merton's lognormal diffusion model for capital gains (see Merton [1971] and Ross [1975]). (There are a number of discrete time models, but with the notable exception of Hakansson's [1971b] and Rubinstein's [1973b], work most of these simply exogenously assume that the market price of risk, $(E_m - E_0)/\sigma_m^2$, remains unchanging over time.) In the Merton model returns are assumed to follow lognormal diffusions, and instantaneous portfolio choice is identical to that of the CAPM. As a consequence, the CAPM is argued to hold intertemporally. In an extension of his original work Merton [1973] showed that Hakansson's [1971a] state dependency problem also could arise in the diffusion model; when the investment opportunity set changes over time in a stochastic fashion, portfolios will be chosen so as to hedge against such movements. This effect, as in the inflation literature, will distort the CAPM relationship and introduce covariances with summary measures of these changes into the expression determining risk premia. In short, the market portfolio is *not* continuously mean variance efficient and the CAPM does not hold in such an environment. The exception to this occurs when preferences in the economy (over terminal wealth or the consumption flow) are represented by a single von-Neumann Morgenstern logarithmic utility function. Now, as Hakansson [1971a] showed, intertemporal returns are really evaluated on an additive basis and such hedging will not occur. While expected utility is affected

13. For an alternative view, see Fama and Schwert [1977].

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by the changing investment set, portfolio behavior is not and the CAPM will hold continuously.

There is, however, an important difficulty with these approaches. There is a strong sense in which individual behavior in the lognormal case, for example, is inherently irrational. At each moment of time the market portfolio is efficient, hence with no stochastic changes in the underlying description of anticipated returns, the same market portfolio (or a deterministically different one) will be chosen at all points of time. But, since the porportions in the market portfolio are total values, with exogenously given supplies, asset prices will also remain unchanged over time (or change deterministically). This means that only dividend uncertainty can exist in equilibrium; capital gains are sure. Or, in the absence of dividend payouts, as in the simpler models, all returns are ex ante viewed as random (lognormally distributed), but ex post prices grow at a certain (identical) rate and returns must be sure. (The model in Ross [1975] notes this difficulty and imposes a myopic rationality criterion, but basically lives with this flaw for the sake of doing some explicit dynamics in a capital goods context.)

The natural solution to this conflict is to model equilibrium in terms of rational price functions. If we require the ex post equilibrium price distribution to be the ex ante one used by consumers and investors, then an equilibrium will be a price function from the states of nature into price space. Leroy [1973] was the first to make this point in a simple two asset version of the CAPM, and Lucas [1975], has examined the same problem in a pure exchange stock market model, but without the CAPM structure.

In a diffusion context if the states of nature are described by a vector, $\mathbf{x}$, that follows an exogenously specified distribution, then all prices will be functionally dependent on $\mathbf{x}$ and their motion will be determined from Ito's lemma. Clearly, by an option hedging argument the number of independent assets cannot exceed the rank of the state space description, $\mathbf{x}$. This leads to a general equilibrium pricing relationship containing an unknown pricing kernel which Garman [1976] uses as the basis for a framework within which intertemporal pricing models can be developed. These points are developed further in Cootner [1977], and in Cox and Ross [1977] which presents a complete rational intertemporal pricing model with a variety of possible specifications including endogenous investment.[14]

Of course, in these models, as in those without a rationality condition (without logarithmic utility), the CAPM no longer holds, but the APT remains valid. As long as the number of assets is larger than the number of factors that are needed to explain the movement in prices at each instant of time, risk premia will be explained by the premia on these factors. More generally, if the number of factors that are essential to the description of all but the idiosyncratic, independent

14. This overstates the case; there is a major problem with all of these models. Outside of the situation where all agents have identical constant relative risk aversion, intertemporal changes in the income distribution will influence the myopic weighting of individual risk aversion coefficients and, thus, affect the intertemporal movement in the risk premia coefficients on states of nature, $\mathbf{x}$. This implies that the distribution itself must be included in the state description and this expands the state description in a troublesome fashion. Some of the existence problems in Section 2 now may become pertinent and, at any rate, explicit analysis becomes quite difficult.

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movement in prices is sufficiently smaller than the set of assets, then instantaneous risk premia on assets will be explained by instantaneous beta coefficients on these factors. At present, this restriction is the only approach that offers testable hypotheses in the intertemporal models.

The exciting work in the area of intertemporal capital models remains to be done. What exactly are the unknown factors that change return anticipations? How will changes in information sets over time affect equilibrium prices and how will these prices and their movements feed back into individual information sets? How are these phenomena to be modelled in terms of observables so as to be tested?

Conclusions

What, then, is the current status of the CAPM? As a theory, its foundations have been broadened in recent years and we are closer to a rather complete understanding of the sufficient conditions for the theory to hold. But, in attempting to extend it beyond its original framework it has proven to be somewhat less robust than might have been hoped for; the mean variance efficiency of the market portfolio is not generally sustained in broader contexts. More importantly, the primary justification for the restriction to mean variance analysis is that the resulting CAPM theory implies potentially testable propositions about asset returns. Roll, however, has argued that previous tests of the theory are defective and, more importantly, that the theory itself is considerably more difficult to test then had been thought. It is not impossible to test the CAPM, but it is ironic that after more than a decade of study, no robust test of a supposedly testable theory exists.

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